JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Sequences and Series of Real Numbers (2005–2014)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
20	99 M	60 Min	Classic Paper Pattern (2

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	20	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q19 • ID: JAM_2014_Q19) MCQ +2 M Neg: −0.67

Q.19 Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series, where $a_n = \frac{(-1)^n n}{2^n}$, $b_n = \frac{(-1)^n}{\log(n+1)}$ for all $n \in \mathbb{N}$. Then

(A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent

(B) $\sum_{n=1}^{\infty} a_n$ is absolutely convergent but $\sum_{n=1}^{\infty} b_n$ is conditionally convergent

(C) $\sum_{n=1}^{\infty} a_n$ is conditionally convergent but $\sum_{n=1}^{\infty} b_n$ is absolutely convergent

(D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are conditionally convergent

Question 2 (JAM 2014 • Q18 • ID: JAM_2014_Q18) MCQ +2 M Neg: −0.67

Q.18 Let $\{x_n\}$ be a sequence of real numbers such that $\lim_{n\to\infty}(x_{n+1}-x_n)=c$, where c is a positive real number. Then the sequence $\left\{\frac{x_n}{n}\right\}$

(A) is NOT bounded

(B) is bounded but NOT convergent

(C) converges to c

(D) converges to 0

Question 3 (JAM 2014 • Q17 • ID: JAM_2014_Q17) MCQ +2 M Neg: −0.67

Q.17 Let $x_n = 2^{2n}\left(1-\cos\left(\frac{1}{2^n}\right)\right)$ for all $n \in \mathbb{N}$. Then the sequence $\{x_n\}$

(A) does NOT converge

(B) converges to 0

(C) converges to $\frac{1}{2}$

(D) converges to $\frac{1}{4}$

Question 4 (JAM 2014 • Q6 • ID: JAM_2014_Q6) MCQ +1 M Neg: −0.33

Q.6 The radius of convergence of the power series $\sum_{n=0}^{\infty} 2^{2n} x^{n^2}$ is

(A) $\frac{1}{4}$ (B) 1 (C) 2 (D) 4

Question 5 (JAM 2013 • Q7 • ID: JAM_2013_Q7) MCQ +2 M Neg: −0.67

Let

$$x_n = \left(1-\frac{1}{3}\right)^2 \left(1-\frac{1}{6}\right)^2 \left(1-\frac{1}{10}\right)^2 \cdots \left(1-\frac{1}{\frac{n(n+1)}{2}}\right)^2, \quad n \geq 2.$$

Then $\lim_{n\to\infty} x_n$ is

(A) $\frac{1}{3}$ (B) $\frac{1}{9}$ (C) $\frac{1}{81}$ (D) 0

Question 6 (JAM 2012 • Q13 • ID: JAM_2012_Q13) MCQ +6 M Neg: −2.0

Q.13 If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges for $x = 3$, then the series $\sum_{n=0}^{\infty} a_n x^n$

(A) converges absolutely for $x = -2$
(B) converges but not absolutely for $x = -1$
(C) converges but not absolutely for $x = 1$
(D) diverges for $x = -2$

Question 7 (JAM 2012 • Q1 • ID: JAM_2012_Q1) **MCQ** +6 M Neg: −2.0

Q.1 Let $\{x_n\}$ be the sequence $+\sqrt{1}, -\sqrt{1}, +\sqrt{2}, -\sqrt{2}, +\sqrt{3}, -\sqrt{3}, +\sqrt{4}, -\sqrt{4}, \ldots$.
If

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n} \text{ for all } n \in \mathbb{N},$$

then the sequence $\{y_n\}$ is

(A) monotonic
(B) NOT bounded
(C) bounded but NOT convergent
(D) convergent

Question 8 (JAM 2011 • Q14 • ID: JAM_2011_Q14) **MCQ** +6 M Neg: −2.0

Q.14 The set of all x at which the power series $\sum_{n=1}^{\infty} \frac{n}{(2n+1)^2}(x-2)^{3n}$ converges is

(A) $[-1, 1)$ (B) $[-1, 1]$ (C) $[1, 3)$ (D) $[1, 3]$

Question 9 (JAM 2011 • Q3 • ID: JAM_2011_Q3) **MCQ** +6 M Neg: −2.0

Q.3 The value of $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{\sqrt{n^2 + kn}}$ is

(A) $2(\sqrt{2}-1)$ (B) $2\sqrt{2}-1$ (C) $2-\sqrt{2}$ (D) $\frac{1}{2}(\sqrt{2}-1)$

Question 10 (JAM 2011 • Q1 • ID: JAM_2011_Q1) **MCQ** +6 M Neg: −2.0

Q.1 Let $a_n = \sum_{k=1}^{n} \frac{n}{n^2 + k}$, for $n \in \mathbb{N}$. Then the sequence $\{a_n\}$ is

(A) Convergent
(B) Bounded but not convergent
(C) Diverges to ∞
(D) Neither bounded nor diverges to ∞

Question 11 (JAM 2010 • Q13 • ID: JAM_2010_Q13) MCQ +6 M Neg: −2.0

Q.13 The radius of convergence of the power series $\sum_{n=0}^{\infty} a_n z^{n^2}$, where $a_0 = 1$, $a_n = 3^{-n} a_{n-1}$ for $n \in \mathbf{N}$, is

(A) 0 (B) $\sqrt{3}$ (C) 3 (D) ∞

Question 12 (JAM 2010 • Q1 • ID: JAM_2010_Q1) MCQ +6 M Neg: −2.0

Q.1 Which of the following conditions does NOT ensure the convergence of a real sequence $\{a_n\}$?

(A) $|a_n - a_{n+1}| \to 0$ as $n \to \infty$

(B) $\sum_{n=1}^{\infty} |a_n - a_{n+1}|$ is convergent

(C) $\sum_{n=1}^{\infty} n\, a_n$ is convergent

(D) The sequences $\{a_{2n}\}$, $\{a_{2n+1}\}$ and $\{a_{3n}\}$ are convergent

Question 13 (JAM 2009 • Q9 • ID: JAM_2009_Q9) MCQ +6 M Neg: −2.0

Q.9 The set of all positive values of a for which the series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \tan^{-1}\left(\frac{1}{n}\right)\right)^{a}$ converges, is

(A) $\left(0, \frac{1}{3}\right]$ (B) $\left(0, \frac{1}{3}\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $\left(\frac{1}{3}, \infty\right)$

Question 14 (JAM 2009 • Q6 • ID: JAM_2009_Q6) MCQ +6 M Neg: −2.0

Q.6 Let $a_n = \begin{cases} \dfrac{1}{3^n} & \text{if } n \text{ is a prime,} \\ \dfrac{1}{4^n} & \text{if } n \text{ is not a prime.} \end{cases}$

Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is

(A) 4 (B) 3 (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Question 15 (JAM 2008 • Q12 • ID: JAM_2008_Q12) MCQ +6 M Neg: −2.0

Q.12 Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers defined as $a_1 = 1$ and for $n \geq 1$,

$a_{n+1} = a_n + (-1)^n 2^{-n}$, $b_n = \dfrac{2a_{n+1} - a_n}{a_n}$. Then

(A) $\{a_n\}$ converges to zero and $\{b_n\}$ is a Cauchy sequence
(B) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is a Cauchy sequence
(C) $\{a_n\}$ converges to zero and $\{b_n\}$ is not a convergent sequence
(D) $\{a_n\}$ converges to a non-zero number and $\{b_n\}$ is not a convergent sequence

Question 16 (JAM 2007 • Q10 • ID: JAM_2007_Q10) MCQ +6 M Neg: −2.0

10. Suppose (c_n) is a sequence of real numbers such that $\lim_{n \to \infty} |c_n|^{1/n}$ exists and is non-zero. If the radius of convergence of the power series $\sum_{n=0}^{\infty} c_n x^n$ is equal to r, then the radius of convergence of the power series $\sum_{n=1}^{\infty} n^2 c_n x^n$ is

(A) less than r
(B) greater than r
(C) equal to r
(D) equal to 0

Question 17 (JAM 2007 • Q7 • ID: JAM_2007_Q7) MCQ +6 M Neg: −2.0

7. Let (a_n) be an increasing sequence of positive real numbers such that the series $\sum_{k=1}^{\infty} a_k$ is divergent. Let $s_n = \sum_{k=1}^{n} a_k$ for $n = 1,2,\ldots$ and $t_n = \sum_{k=2}^{n} \dfrac{a_k}{s_{k-1} s_k}$ for $n = 2,3,\ldots$. Then $\lim_{n \to \infty} t_n$ is equal to

(A) $1/a_1$
(B) 0
(C) $1/(a_1 + a_2)$
(D) $a_1 + a_2$

Question 18 (JAM 2006 • Q1 • ID: JAM_2006_Q1) MCQ +6 M Neg: −2.0

1. $\lim_{n\to\infty} \dfrac{2^{n+1}+3^{n+1}}{2^n+3^n}$ equals

(A) 3
(B) 2
(C) 1
(D) 0

Question 19 (JAM 2005 • Q14 • ID: JAM_2005_Q14) MCQ +6 M Neg: −2.0

14. Let $f:(-1,1) \longrightarrow \boldsymbol{R}$ be such that $f^{(n)}(x)$ exists and $\left|f^{(n)}(x)\right| \le 1$ for every $n \ge 1$ and for every $x \in (-1,1)$. Then f has a convergent power series expansion in a neighbourhood of

(A) every $x \in (-1,1)$

(B) every $x \in \left(\frac{-1}{2}, 0\right)$ only

(C) no $x \in (-1,1)$

(D) every $x \in \left(0, \frac{1}{2}\right)$ only

Question 20 (JAM 2005 • Q1 • ID: JAM_2005_Q1) MCQ +6 M Neg: −2.0

1. Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences of real numbers such that $b_n = a_{2n}$ and $c_n = a_{2n+1}$. Then $\{a_n\}$ is convergent

(A) implies $\{b_n\}$ is convergent but $\{c_n\}$ need not be convergent

(B) implies $\{c_n\}$ is convergent but $\{b_n\}$ neet not be convergent

(C) implies both $\{b_n\}$ and $\{c_n\}$ are convergent

(D) if both $\{b_n\}$ and $\{c_n\}$ are convergent

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Sequences and Series of Real Numbers (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	**B**
2	MCQ	+2	**C**
3	MCQ	+2	**C**
4	MCQ	+1	**B**
5	MCQ	+2	**B**
6	MCQ	+6	**A**
7	MCQ	+6	**D**
8	MCQ	+6	**C**
9	MCQ	+6	**A**
10	MCQ	+6	**A**

Q. No.	Type	Marks	Official Key
11	MCQ	+6	**B**
12	MCQ	+6	**A**
13	MCQ	+6	**C**
14	MCQ	+6	**B**
15	MCQ	+6	**B**
16	MCQ	+6	**C**
17	MCQ	+6	**A**
18	MCQ	+6	**A**
19	MCQ	+6	**A**
20	MCQ	+6	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.